
ROBERT ANGELO BORRELLI
ASSOCIATE PROFESSOR
University of Idaho · Idaho Falls Center for Higher Education
Department of Nuclear Engineering and Industrial Management
Center for Advanced Energy Studies · 995 MK Simpson Boulevard · Idaho Falls ID 83401
rborrelli@uidaho.edu · @TheDoctorRAB

EDUCATION

Doctor of Philosophy – Nuclear Engineering 2006
University of California – Berkeley
Radionuclide transport modeling with bentonite extrusion

Master of Science – Civil & Environmental Engineering 1999
Worcester Polytechnic Institute
Characterization of background radiation in the environment

Bachelor of Science, Mechanical Engineering with high distinction 1996
Worcester Polytechnic Institute
Capstone – Real time PLC-based reactivity modeling by inverse point kinetics

RESEARCH EXPERIENCE

University of Idaho · Idaho Falls Center for Higher Education 2015–
Associate Professor · Nuclear Engineering & Industrial Management 2021–
Assistant Professor · Nuclear Engineering & Industrial Management 2015–21
Affiliate Faculty – Electrical & Computer Engineering · University of Idaho 2025–
Coordinator – Nuclear Technology Management Certificate 2023–
Affiliate – Energy Policy Institute · Boise State University 2019–
Coordinator – Nuclear Decommissioning & Used Fuel Management Certificate 2019–
State of Idaho Professional Engineer – Faculty Restricted 2019–
Coordinator – Nuclear Criticality Safety Certificate 2015–

Postdoctorate Researcher 2009-12
University of California-Berkeley · Nuclear Engineering
Safeguardability methodology for remotely-handled nuclear materials facilities

Research Associate 2007–09
University of Tokyo · Nuclear Engineering & Management
Mathematical modeling for mass transport in the engineered barrier system of a high-level nuclear waste repository

Postdoctorate Researcher 2007
University of California-Berkeley · Nuclear Engineering
Mathematical modeling for mass transport in the engineered barrier system of a high-level nuclear waste repository

Doctoral Candidate 2005-06
University of California-Berkeley · Nuclear Engineering
Derived a two-phase, mass transport model for radionuclides in a porous medium with bentonite extrusion model in a planar fracture to assess the potential to confine radionuclides.

Intern, Earth Sciences Division
Lawrence Berkeley National Laboratory
Data analysis for chaotic modeling of water flow in the unsaturated zone

2001–02

RELATED PROFESSIONAL EXPERIENCE

Assistant Radiation Safety Officer 1996–99
Leslie C. Wilbur Nuclear Reactor Facility
Worcester Polytechnic Institute
Emergency response for incidents involving radioactive material; Records maintenance of radiation exposures to campus personnel, etc.

Senior Nuclear Reactor Operator #70145 1994–99
Leslie C. Wilbur Nuclear Reactor Facility
Worcester Polytechnic Institute
Emergency response for incidents involving radioactive material; Records maintenance of radiation exposures to campus personnel, etc.

Major Qualifying Project 1995–96
Leslie C. Wilbur Nuclear Reactor Facility
Worcester Polytechnic Institute
The Major Qualifying Project synthesizes previous undergraduate studies to solve problems in the major field and communicate results. For this project, a logic algorithm was developed for reactivity derived from point kinetics equations and a Programmable Logic Controller was modified for real time data collection during nuclear reactor operation for use with experimental coursework and maintenance procedures.

GRANTS AND CONTRACTS AWARDED

- (1) Kathleen Araújo (PI), Cassie Koerner (co-PI) – Boise State University/Energy Policy Institute, R. A. Borrelli (co-PI) – University of Idaho, Idaho Falls Center for Higher Education, Richelle Johnson (co-PI) – University of Alaska, Anchorage, Stanley A. Orrell (Senior Personnel) – Consultant, Integrated Waste Management, Christi Bell (Senior Personnel) - University of Alaska, Anchorage/Center for Economic Development. Common Ground 2050: Public Engagement and Ambition as Preconditions for National Nuclear Readiness with Integrated Waste Management. NEUP 25-34204. **\$1,000,000**. 2026.08.01 - 2029.07.31.
- (2) Sean McBride (PI) - Idaho State University, Dakota Roberson (co-PI), R. A. Borrelli (co-PI), Constantinos Kolias (co-PI) - University of Idaho. Industrial Cyber Security Research and Training Laboratory. National Institute of Standards and Technology. **\$2,875,000**. 2024.08.01 - 2026.07.31. [non-competitive]
- (3) Michael Haney (PI), R. A. Borrelli (co-PI), Dakota Roberson (co-PI), Constantinos Kolias (co-PI) - University of Idaho, Ben Lampe (co-PI), Sean McBride (co-PI) - Idaho State University. Secure Cyberspace and Resilient Industrial Systems Workforce Development. Idaho Global Entrepreneurial Mission Initiative • Higher Education Research Council. **\$700,000**. 2024.07.01 - 2025.06.30. [Borrelli PI 2024.07.01]
- (4) Kathleen Araújo (PI), Cassie Koerner (co-PI) - Boise State University, Stephanie Malin (co-PI) - Colorado State University, Daniel Cardenas (co-PI) - National Tribal Energy Association, R. A. Borrelli (co-PI) - University of Idaho, Weston Eaton (co-PI), Temple Stoellinger (Senior Personnel), Steven Smutko (Senior Personnel), Rachael Budowle (Senior Personnel) - University of Wyoming, Majia Nadesan (co-PI) - Arizona State University, Julia Haggerty (co-PI), Lee Spangler (Senior Personnel) - Montana State University, Denia Djokić (co-PI) University of Michigan, Sarah Robey (co-PI) - Idaho State University. Common ground: Legitimacy in consent-based siting for interim nuclear waste storage. United States Department of Energy Consent-Based Siting for Interim Storage Program - Community Engagement Opportunities. **\$2,000,000**. 2023.08.01 - 2025.07.31.
- (5) Andrew Kliskey (PI) - Idaho EPSCoR Director, Karla Eitel (co-PI), Alistair Smith (co-PI) - Uni-

versity of Idaho, Donna Lybecker (co-PI) - Idaho State University, Kathleen Araújo (co-PI) Boise State University. [RII Track-1: Idaho Community-engaged Resilience for Energy-Water Systems \(I-CREWS\)](#). National Science Foundation EPSCoR. **\$24,000,000**. 2023.09.01 - 2028.08.31.^{1,2} [*Modeling Group Lead 2025.09.01*] [*Treasure Valley Site Planning Team 2026.01.16*]

- (6) Michael Haney (PI), R. A. Borrelli (co-PI), Dakota Roberson (co-PI), Constantinos Kolias (co-PI) - University of Idaho, Ben Lampe (co-PI), Sean McBride (co-PI) - Idaho State University. Secure Cyberspace and Resilient Industrial Systems Workforce Development. Idaho Global Entrepreneurial Mission (IGEM) – Higher Education Research Council **\$700,000**. 2023.07.01 - 2024.06.30.
- (7) Michael Haney (PI), R. A. Borrelli (co-PI), Dakota Roberson (co-PI), Constantinos Kolias (co-PI) - University of Idaho, Ben Lampe (co-PI), Sean McBride (co-PI) - Idaho State University. Secure Cyberspace and Resilient Industrial Systems Workforce Development. Idaho Global Entrepreneurial Mission (IGEM) – Higher Education Research Council **\$693,000**. 2022.07.01 - 2023.06.30.
- (8) R. A. Borrelli (PI), Michael Haney (co-PI) - University of Idaho. Cyber-informed design, education, and training for cyberthreat resiliency with real-time nuclear reactor simulation. University of Idaho. Operation: Resubmission Support. **\$34,122**. 2022.04.30 - 2022.09.30.
- (9) Thomas A. Ulrich (PI) - Idaho National Laboratory, R. A. Borrelli (co-PI) - University of Idaho. User evaluation of the NuScale simulator at the Center for Advanced Energy Studies. CAES programmatic funding. **\$50,000**. 2022.03.01 - 2022.09.30.
- (10) R. A. Borrelli (PI) - University of Idaho, Dennis D. Keiser, Jr., (co-PI) - Idaho National Laboratory. Graduate Research Assistantship: Connecting U-Mo Fuel Processing, Microstructure, and Irradiation Performance. **\$23,823**. 2021.06.01-2022.01.31.
- (11) R. A. Borrelli (PI), Jason Barnes (Senior Adviser) - University of Idaho. Experimental determination of interactions between the radiation fields of Dragonfly’s MMRTG and Titan’s environment. Idaho NASA EPSCoR Research Initiation Grant. **\$82,962**. 2021.05.01 - 2022.04.30.
- (12) R. A. Borrelli (PI) - University of Idaho. Online educational resources for nuclear engineering courses. University of Idaho Think Open Fellowship. **\$1200**. 2021.01.01 - 2021.05.31.
- (13) Lee Ostrom (PI), Richard N. Christensen, R. A. Borrelli, Haiyan Zhao (co-PIs) - University of Idaho. ORED Fall 2019 EIS: Portable XFR for use in supporting material research. ORED Equipment and Infrastructure Support. **\$40,000**. 2019.12.01 - 2020.11.30
- (14) R. A. Borrelli (PI) - University of Idaho, Mark. D. DeHart (co-PI) - Idaho National Laboratory. Application and enhancement of MAMMOTH depletion capabilities. **\$33,521**. 2020.01.13 - 2020.12.31.
- (15) Richard N. Christensen (PI), R. A. Borrelli, Michael G. McKellar, Michael Haney, David Arcilesi (co-PIs) - University of Idaho, Richard Jacobson (co-PI) Idaho State University. NuScale Simulator at the Center for Advanced Energy Studies. United States Department of Energy Scientific Infrastructure Support for Consolidated Innovative Nuclear Research. **\$321,525**. 2019.10.01 - 2022.09.30. [*PI - NuScale Simulator Laboratory - 2022.01.07*]
- (16) R. A. Borrelli (PI) - University of Idaho, Dennis D. Keiser, Jr., (co-PI) - Idaho National Laboratory. Graduate Research Assistantship: Connecting U-Mo Fuel Processing, Microstructure, and Irradiation Performance. **\$127,866**. 2018.10.01-2021.05.31.
- (17) R. A. Borrelli (PI), Richard N. Christensen (co-PI) - University of Idaho, Brian J. Jaques (co-PI) - Boise State University, Piyush Sabharwall (co-PI) - Idaho National Laboratory, Mark Delligatti (co-PI) - Table Rock, LLC, Sakae Casting USA, LLC (co-PI). Modeling and design of borated aluminium cask for used fuel cooling. Idaho Global Entrepreneurial Mission (IGEM) - Idaho Commerce. **\$237,898**. 2018.01.01-2019.05.31.
- (18) R. A. Borrelli (PI) - University of Idaho, Dennis D. Keiser, Jr., (co-PI) - Idaho National Laboratory. Graduate Research Assistantship: Connecting U-Mo Fuel Processing, Microstructure, and Irradiation Performance. **\$36,180**. 2017.11.01-2018.05.31.

¹Technical Proposal Writing Team.

²University of Idaho Research Team.

- (19) R. A. Borrelli (PI), Lee Ostrom (Senior Advisor) - University of Idaho, Stephen G. Johnson (Senior Advisor) - Idaho National Laboratory. Performance assessment of americium as fuel in radioisotope thermoelectric generators for deep space exploration. Idaho NASA EPSCoR Research Initiation Grant. **\$55,000**. 2017.08.01-2018.04.30.
- (20) Kelley M. Verner (PI), R. A. Borrelli, Marc T. Skinner, Emma K. Redfoot, Jieun Lee, Seth Dustin, John Peterson (co-PIs) - University of Idaho. Increasing the Go-on Rate in Southeast Idaho Through the Nexus of Food, Energy, and Water. University of Idaho Vandals Big Ideas Project. **\$23,984**. 2017.07.01-2018.06.30.
- (21) R. A. Borrelli (PI) - University of Idaho, Jason Hales (co-PI) - Idaho National Laboratory. Graduate Research Assistantship: Idaho National Laboratory Code Documentation. **\$35,435**. 2016.10.01-2017.06.30.
- (22) Vivek Utgikar (PI), Fatih Aydogan, Krishnan Raja, Raghunath Kanakala, R. A. Borrelli, Haiyan Zhao, Matthew Swenson (co-PIs) - University of Idaho. University of Idaho Nuclear Engineering Faculty Development Program. United States Regulatory Commission Faculty Development Grant. **\$434,048**. 2015.09.29 - 2019.09.30.

Major Equipment Acquisitions

Western Services Corporation Pressurized Water Reactor Nuclear Power Plant Simulator.

REFEREED JOURNAL PUBLICATIONS

- (1) K. Araujo, A. Kouts, R. A. Borrelli, A. Miller, S. E. Robey, J. Haggerty (2026). Procedural Trust in Nuclear Waste Siting: Lessons from Yucca Mountain, WIPP, and a Goshute Initiative. *Nature Sustainability*, [under review].
- (2) Cassie Koerner, R. A. Borrelli (2026). A Collaborative-Based Siting case study: Yucca Mountain siting – The Willrich report predictions. *Nuclear Technology*, [under review].
- (3) Haruko M. Wainwright, Shaheen A. Dewji, John McCloy, Brian A. Powell, R. A. Borrelli, Massimiliano Fratoni, Jeffrey Wang, Liam Hines, Dauren Sarsenbayev, Craig H. Benson (2026). A New Paradigm for High-Level Radioactive Waste Disposal: Intrinsic Radionuclide Properties and Comparative Hazard. *Proceedings of the National Academy of Sciences of the United States of America*, [under review].
- (4) David Haar, R. A. Borrelli (2026). The Elimination of High Dose-Rate Neutron Activation Products within Nuclear Power Plant Structural Materials. *Nuclear Engineering and Design*, [under review].
- (5) Tollan Berhanu, Hunter Squires, Braxton Marlatt, Scott Anderson, Benton Wilson, R. A. Borrelli, Constantinos Kolias (2026). AE³GIS — An Agile Emulated Educational Environment for Guided Industrial Security Training. *Future Internet*, 10.3390/fi18030166.
- (6) Ethan Bauer, Tommy Holschuh, R. A. Borrelli (2026). Calibration Source Selection and Covariance Uncertainty Propagation in HPGe Efficiency Calibration. *Radiation Measurements*, [under review].
- (7) Ethan Bauer, Tommy Holschuh, R. A. Borrelli (2025). Evaluation of Simulated HPGe Detector Efficiencies in OpenMC Compared to MCNP. *Nuclear Engineering and Technology*, 104167.
- (8) Koffi Anderson Koffi, Kyle Lucke, Elijah Danquah Darko, Tollan Berhanu, R. A. Borrelli, Constantinos Kolias (2025). AgentRed: Towards an Agent-Based Approach to Network Red Team Assessment Automation. *Algorithms Artificial Intelligence in Modern Cybersecurity: Changes, Applications and Challenges*, 10.3390/a19010043.
- (9) Nathan Manwaring, Matt Johnson, R. A. Borrelli (2024). At-power Subcritical Multiplication in the Advanced Test Reactor during Nuclear Requalification Testing. *Nuclear Engineering and Design* 426, 113399.
- (10) Olin Calvin, Barry D. Ganapol, R. A. Borrelli (2023). Introduction of the adding and doubling method for solving Bateman equations for nuclear fuel depletion. *Nuclear Science and Engineering* 197, 558.
- (11) Pedro Mena, R. A. Borrelli, Leslie Kerby (2023). Detecting Anomalies in Simulated Nuclear Data

- using Autoencoders. *Nuclear Technology* 210, 112.
- (12) Sam J. Root, Porter Throckmorton, Jonathan Tacke, Jacob Benjamin, Michael Haney, R. A. Borrelli (2023). Cyber Hardening of Nuclear Power Plants with Real-time Nuclear Reactor Operation — 1. Preliminary Operational Testing. *Progress in Nuclear Energy* 162, 104742.
 - (13) Sam J. Root, Haiyan Zhao, R. A. Borrelli, Michael G. McKellar (2023). Thermodynamic Analysis on Xenon Stripping to Shorten Restart Time in Molten Salt Microreactors. *Nuclear Engineering and Design* 414, 112606.
 - (14) Haruko Wainwright, Brian Powell, Megan Hoover, Dinara Ermakova, R. A. Borrelli, Christina Leggett, Milos Atz, Carol Eddy-Dilek, William Roy, Emily Stein, Kathryn Higley, Sheldon Landsberger, Rod Ewing (2023). Nuclear Waste Educator's Workshop: What and How Do We Teach About Nuclear Waste? *Journal of Environmental Radioactivity* 270, 107288.
 - (15) Teyen Widdicombe, R. A. Borrelli (2023). Experimental Determination of Interaction Between the Radiation Fields of Dragonfly's MMRTG and Titan's Environment II: Gamma Induced Atmospheric Conductivity. *Acta Astronautica* 208, 91.
 - (16) Joseph Christensen, R. A. Borrelli (2022). Evaluations of the effect of heterogeneity in HALEU systems using modified critical benchmarks. *Nuclear Science and Engineering* 196, 1333.
 - (17) Nathan Manwaring, R. A. Borrelli (2022). At-power subcritical multiplication in the Advanced Test Reactor. *Nuclear Engineering and Design* 401, 112040.
 - (18) Pedro Mena, R. A. Borrelli, Leslie Kerby (2022). Survey of markets for nuclear power in Western North America. *International Journal of Energy, Environment, and Economics* 29, 17.
 - (19) Pedro Mena, R. A. Borrelli, Leslie Kerby (2022). Expanded Analysis of Machine Learning Models for Nuclear Transient Identification Using TPOT. *Nuclear Engineering and Design* 390, 111694.
 - (20) Emma K. Redfoot, Kelley M. Verner, R. A. Borrelli (2022). Applying analytic hierarchy process to industrial process design in a nuclear renewable hybrid energy system. *Progress in Nuclear Energy* 145, 104083.
 - (21) Emma K. Redfoot, Michael G. McKellar, R. A. Borrelli (2022). Allocating heat and electricity in an integrated energy system coupled with a water purification system. *Nuclear Engineering and Design* 397, 111902.
 - (22) Teyen Widdicombe, R. A. Borrelli (2022). Experimental Determination of Interaction Between the Radiation Fields of Dragonfly's MMRTG and Titan's Environment I: Neutron results. *Acta Astronautica*, 10.1016/j.actaastro.2022.08.048.
 - (23) J. Seth Dustin, R. A. Borrelli (2021). Assessment of alternative radionuclides for use in a radioisotope thermoelectric generator. *Nuclear Engineering and Design* 385, 111475.
 - (24) J. Seth Dustin, R. A. Borrelli (2021). Modeling of Am-241 as an alternative fuel source in a radioisotope thermoelectric generator. *Nuclear Engineering and Design* 385, 111495.
 - (25) Pedro Mena, R. A. Borrelli, Leslie Kerby (2021). Nuclear reactor transient diagnostics using classification and AutoML. *Nuclear Technology* 208, 232.
 - (26) Jonathan Tacke, R. A. Borrelli, Dakota Roberson (2021). Advanced frequency-domain compensator design for subsystems within a nuclear generating station. *Progress in Nuclear Energy* 140, 103914.
 - (27) Teyen Widdicombe, R. A. Borrelli (2021). MCNP modelling of radiation effects of the Dragonfly mission's RTG on Titan II: Atmospheric ionization effects. *Acta Astronautica* 186, 517.
 - (28) Teyen Widdicombe, R. A. Borrelli (2021). MCNP modelling of radiation effects of the Dragonfly mission's RTG on Titan. *Acta Astronautica* 183, 363.
 - (29) R. A. Borrelli, Mark S. Delligatti, Brenden J. Heidrich (2020). Borated aluminum cask design for onsite intermediate storage - neutronics design and certification analysis. *Nuclear Engineering and Design* 363, 110666.
 - (30) John P. Carter, R. A. Borrelli (2020). Integral molten salt reactor neutron physics study using Monte Carlo N-particle code. *Nuclear Engineering and Design* 365, 110718.

- (31) John P. Carter, R. A. Borrelli (2020). Integral molten salt reactor temperature sensitivities using Serpent target motion sampling. *Nuclear Engineering and Design* 369, 110863.
- (32) Joseph Christensen, R. A. Borrelli (2020). Nuclear criticality safety aspects for the future of HALEU: Evaluating heterogeneity in intermediate-enrichment uranium using critical benchmark experiments. *Nuclear Science and Engineering*, 10.1080/00295639.2020.1819143.
- (33) Joseph Christensen, R. A. Borrelli (2020). Parametric study of minimum critical volume for high-assay low-enriched uranium (20%) in spherical geometry against particle size. *Nuclear Science and Engineering* 196, 98.
- (34) Kaelee A. Novich, Samuel V. Pedersen, R. A. Borrelli, Richard N. Christensen, Brian J. Jaques (2020). Synthesis of boron carbide reinforced aluminum castings through mechanical stir casting. *Journal of Composite Materials* 55, 10.1177/0021998320987597.
- (35) Jieun Lee, R. A. Borrelli (2019). Sensitivity analysis and application of advanced nuclear accounting methodologies on the high reliability safeguards model: Use of discrete event simulation for material throughput. *Nuclear Engineering and Design* 345, 183.
- (36) Jieun Lee, Amey Shigrekar, R. A. Borrelli (2019). Hazard and operability analysis of a pyroprocessing facility. *Nuclear Engineering and Design* 348, 131.
- (37) John Peterson, Michael Haney, R. A. Borrelli (2019). An overview of methodologies for cybersecurity vulnerability assessment conducted in nuclear power plants. *Nuclear Engineering and Design* 346, 75.
- (38) Emma K. Redfoot, R. A. Borrelli (2018). Analysis of nuclear renewable hybrid energy systems modeling and nuclear fuel cycle simulators. *Nuclear Technology* 204, 249.
- (39) R. A. Borrelli, Joonhong Ahn, Yongsoo Hwang (2017). Approaches to a practical systems assessment for safeguardability of the advanced nuclear fuel cycle. *Nuclear Technology* 197, 248.
- (40) Jieun Lee, Malachi Tolman, R. A. Borrelli (2017). High reliability safeguards approach to remotely handled nuclear processing facilities: Use of discrete event simulation for material throughput for fuel fabrication. *Nuclear Engineering and Design* 324, 54.
- (41) R. A. Borrelli (2016). Functional components for a design strategy: Hot cell shielding in the high reliability safeguards methodology. *Nuclear Engineering and Design* 305, 18.
- (42) R. A. Borrelli (2014). A high reliability safeguards approach for safeguardability of remotely-handled nuclear facilities: 1. Functional components to system design. *Journal of Nuclear Materials Management* XLII, 4.
- (43) R. A. Borrelli (2014). A high reliability safeguards approach for safeguardability of remotely-handled nuclear facilities: 2. A risk-informed approach for safeguards. *Journal of Nuclear Materials Management* XLII, 27.
- (44) R. A. Borrelli (2014). Use of curium neutron flux from head-end pyroprocessing subsystems for the high reliability safeguards methodology. *Nuclear Engineering Design* 277, 166.
- (45) R. A. Borrelli (2013). Use of curium spontaneous fission neutrons for safeguardability of remotely-handled nuclear facilities: Fuel fabrication in pyroprocessing. *Nuclear Engineering and Design* 260, 64.
- (46) R. A. Borrelli, Olivier Thivent, Joonhong Ahn (2013). Parametric studies on confinement of radionuclides in the excavated damaged zone due to bentonite type and temperature change. *Physics and Chemistry of the Earth* 65, 32.
- (47) R. A. Borrelli, Olivier Thivent, Joonhong Ahn (2011). Impacts of elevated temperatures on bentonite extrusion and radionuclide transport in the excavated damaged zone. *Nuclear Technology* 174, 94.
- (48) R. A. Borrelli, Joonhong Ahn (2008). Radionuclide transport in a water-saturated planar fracture with bentonite extrusion. *Nuclear Technology* 164, 442.
- (49) R. A. Borrelli, Joonhong Ahn (2008). Numerical modeling of bentonite extrusion and radionuclide migration in a saturated planar fracture. *Physics and Chemistry of the Earth* 33, S131.

REFEREED CONFERENCE PROCEEDINGS

- (1) R. A. Borrelli, Denia Djokić (2026). A Collaborative Based Siting Case Study for Nuclear Siting Tools. Denver, Colorado: Proc., American Nuclear Society Annual Meeting.
- (2) Haruko M. Wainwright, Shaheen A. Dewji, John McCloy, Brian A. Powell, R. A. Borrelli, Massimiliano Fratoni, Jeffrey Wang, Liam Hines, Dauren Sarsenbayev, Craig H. Benson (2026). A Paradigm Shift for High-Level Radioactive Waste Disposal: Radionuclides Intrinsic Properties for Long-term Safety. Denver, Colorado: Proc., American Nuclear Society Annual Meeting.
- (3) Cassie Koerner, R. A. Borrelli (2025). A Collaborative-Based Siting case study Yucca Mountain siting: The Willrich report predictions. Washington, D. C.: Proc., International High Level Radioactive Waste Management.
- (4) Eugene Balsmeier, R. A. Borrelli, Craig L. Porter (2025). Radiological Assessment for Sub-seabed Repositories. Phoenix, Arizona: Proc., Waste Management Symposia.
- (5) R. A. Borrelli, Kathleen Araújo, Cassie Koerner, Denia Djokić (2024). Consent based siting for Spent Nuclear Fuel – The Common Ground Consortium Focus on Research and Public Conversations. Las Vegas, Nevada: Proc., American Nuclear Society Annual Meeting.
- (6) Haruko Wainwright, Brian Powell, Megan Hoover, R. A. Borrelli, Carol Eddy-Dilek, Sheldon Landsberger, Rod Ewing (2023). Nuclear Waste Educator’s Workshop: What and How Do We Teach About Nuclear Waste? Indianapolis, Indiana: Proc., American Nuclear Society Annual Meeting.
- (7) Jonathon Wheelwright, R. A. Borrelli (2023). A Regulatory Approach to Nuclear Reactor Transportation. Washington, D. C.: Proc., American Nuclear Society Annual Meeting.
- (8) Sam J. Root, Porter Throckmorton, Michael Haney, R. A. Borrelli (2022). Simulated boron shimming cyber-attack on pressurized water reactor. Phoenix, Arizona: Proc., American Nuclear Society Winter Meeting.
- (9) Michael Haney, Jacob Benjamin, R. A. Borrelli (2021). Cyberweapon nonproliferation controls for the virtual battlefield - Applying the nuclear nonproliferation regime to an unseen enemy. Washington, D. C.: Proc., American Nuclear Society Winter Meeting.
- (10) James Richards, R. A. Borrelli (2020). Development of a tool for evaluating nuclear integrated energy systems profitability for existing reactors. [*virtual*]: Proc., American Nuclear Society Virtual Winter Meeting.
- (11) R. A. Borrelli (2019). Updates on borated aluminum cask design for onsite used fuel storage. Minneapolis, Minnesota: Proc., American Nuclear Society Annual Meeting.
- (12) Joseph Christensen, R. A. Borrelli, Charles S. Henkel, Brian Matthews (2019). Minimum accident of concern for uranyl sulfate solutions. Minneapolis, Minnesota: Proc., American Nuclear Society Annual Meeting.
- (13) Constantinos Kolias, R. A. Borrelli, Daniel Barbara, Angelos Stavrou (2019). Malware detection in critical infrastructures by using the electromagnetic emissions of plcs. Washington, D. C.: Proc., American Nuclear Society Winter Meeting.
- (14) Trevor MacLean, Michael Haney, R. A. Borrelli (2019). Cybersecurity modeling of non-critical nuclear power plant instrumentation. Arlington, Virginia: Proc., Thirteenth Annual IFIP WG 11.10 International Conference on Critical Infrastructure Protection.
- (15) John Peterson, Michael Haney, R. A. Borrelli (2019). An overview of methodologies for cybersecurity vulnerability assessments conducted at nuclear power plants. Washington, D. C.: Proc., American Nuclear Society Winter Meeting.
- (16) R. A. Borrelli, J. Seth Dustin, Samuel V. Pedersen, Brian J. Jaques (2018). Design of a borated aluminum cask design for onsite used fuel storage. Philadelphia, Pennsylvania: Proc., American Nuclear Society Annual Meeting.
- (17) R. A. Borrelli, Mark S. Delligatti (2018). Regulatory licensing pathway for a borated aluminum cask design for onsite used fuel storage. Philadelphia, Pennsylvania: Proc., American Nuclear Society Annual Meeting.

- (18) J. Seth Dustin, R. A. Borrelli (2018). Preliminary performance assessment of americium-241 as fuel in radioisotope thermoelectric generators for deep space exploration. Las Vegas, Nevada: Proc., ANS NETS 2018 - Nuclear and Emerging Technologies for Space.
- (19) Jieun Lee, R. A. Borrelli (2018). Upgrades on high reliability safeguards model for material throughput in fuel fabrication. Philadelphia, Pennsylvania: Proc., American Nuclear Society Annual Meeting.
- (20) R. A. Borrelli, Malachi Tolman, Jieun Lee (2017). Preliminary analysis of facility design for pyroprocessing safeguardability. San Francisco, California: Proc., American Nuclear Society Annual Meeting.
- (21) Jieun Lee, R. A. Borrelli (2017). Hazard and operability analysis of a pyroprocessing facility. Washington, D. C.: Proc., American Nuclear Society Winter Meeting.
- (22) John Peterson, Michael A. Haney, R. A. Borrelli (2017). Cybersecurity vulnerability assessment methodologies for nuclear power plants. San Francisco, California: Proc., American Nuclear Society Annual Meeting.
- (23) Emma K. Redfoot, Kelley M. Verner, R. A. Borrelli (2017). Applying preliminary hazard analysis and analytic hierarchy process to a nuclear renewable hybrid energy system. Washington, D. C.: Proc., American Nuclear Society Winter Meeting.
- (24) Emma K. Redfoot, R. A. Borrelli (2017). Nuclear fuel cycle simulator as a means to model a nuclear hybrid energy system. San Francisco, California: Proc., American Nuclear Society Annual Meeting.
- (25) R. A. Borrelli, Malachi Tolman (2016). Development of a discrete event simulation model for pyroprocessing safeguardability. Las Vegas, Nevada: Proc., American Nuclear Society Winter Meeting.
- (26) R. A. Borrelli (2013). Summary of modeling studies in the high reliability safeguards approach. Washington, D. C.: Proc., American Nuclear Society Winter Meeting.
- (27) R. A. Borrelli (2013). Applying curium spontaneous fission neutrons in the high reliability safeguards approach for remotely-handled nuclear facilities. Salt Lake City, Utah: Proc., Global 2013 - International Nuclear Fuel Cycle Conference.
- (28) R. A. Borrelli (2012). Applying curium spontaneous fission neutrons in the high reliability safeguards approach for remotely-handled nuclear facilities. Savannah, Georgia: Proc., 9th International Conference on Facility Operations-Safeguards Interface.
- (29) R. A. Borrelli (2011). Role of neutron detection in the high reliability safeguards approach. Albuquerque, New Mexico: Proc., International Conf., High-Level Radioactive Waste Management.
- (30) R. A. Borrelli, Lance Kim, Edward D. Blandford, Yongsoo Hwang, Eung Ho Kim, Per F. Peterson (2010). High reliability safeguards for remote-handled nuclear materials. San Diego, California: Proc., 2010 International Congress on Advances in Nuclear Power Plants (ICAPP '2010).
- (31) R. A. Borrelli, Joonhong Ahn (2010). Parametric studies on confinement of radionuclides in the excavated damaged zone due to bentonite type and temperature change. Nantes, France: Proc., Clays in Natural & Engineered Barriers for Radioactive Waste Confinement, 4th International Meeting.
- (32) R. A. Borrelli, Olivier Thivent, Joonhong Ahn (2009). Effect of temperature change on bentonite extrusion and radionuclide migration in the excavated damaged zone. Luxembourg: Proc., Impact of Thermo-Hydro-Mechanical-Chemical Processes on the Safety of Underground Repositories Conference.
- (33) R. A. Borrelli, Olivier Thivent, Joonhong Ahn (2009). Parametric studies for temperature changes of bentonite extrusion and impacts on radionuclide migration in the excavated damaged zone. Kennewick, Washington: Proc., 12th International Conference on the Chemistry and Migration Behaviour of Actinides and Fission Products in the Geosphere.
- (34) R. A. Borrelli, Joonhong Ahn (2008). Impact of bentonite extrusion on radionuclide migration in a saturated planar fracture. Osaka, Japan: Proc., 2008 Annual Meeting of the Atomic Energy Society of Japan.
- (35) R. A. Borrelli, Joonhong Ahn (2007). Impact of bentonite extrusion on radionuclide in a saturated

planar fracture. Lille, France: Proc., Clays in Natural & Engineered Barriers for Radioactive Waste Confinement, 3rd International Meeting.

- (36) R. A. Borrelli, Joonhong Ahn (2007). Radionuclide transport in a water-saturated planar fracture with bentonite extrusion. J. Grunlan, D. Bhattacharyya, E. Marand, O. Regev, A. Balazs (eds.), 1006E. San Francisco, California: Proc., Transport Behavior in Heterogeneous Polymeric Materials and Composites, Mater. Res. Soc. Symp.
- (37) R. A. Borrelli, Joonhong Ahn (2006). Bentonite extrusion in planar fracture based on net water flow. Las Vegas, Nevada: Proc., International Conf., High-Level Radioactive Waste Management.
- (38) Leo M. Bobek, R. A. Borrelli (1996). Plc-based reactivity measurements using inverse point kinetics. Reno, Nevada: Proc., American Nuclear Society Annual Meeting.

BOOK CONTRIBUTIONS

- (1) Trevor MacLean, Robert Borrelli, Michael Haney (2019). Cybersecurity modeling of non-critical nuclear power plant digital instrumentation. In: Jason Staggs, Sujeet Sheno, eds., Critical Infrastructure Protection XIII, chap. 15. Springer, p. 277.
- (2) R. A. Borrelli (2015). Reflections on developing an identity for the third generation nuclear engineer in the post-Fukushima society. In: Joonhong Ahn, Cathryn Carson, Mikael Jensen, Kohta Juraku, Shinya Nagasaki, Satoru Tanaka, eds., Reflections on the Fukushima Daiichi nuclear accident: Toward social scientific literacy and engineering resilience. Springer Cham. ISBN 978-3-319-12089-8, p. 353.

NON-REFEREED TECHNICAL REPORTS

- (1) Haruko Wainwright, Brian Powell, Megan Hoover, Dinara Ermakova, R. A. Borrelli, Christina Leggett, Milos Atz, Carol Eddy-Dilek, William Roy, Emily Stein, Kathryn Higley, Sheldon Landsberger, Rod Ewing (2023). Nuclear Waste Educator's Workshop: What and How Do We Teach About Nuclear Waste? Nuclear Waste Educator's Working Group.
- (2) Kathleen Araújo, Brittany Brand, R. A. Borrelli, Vince Bowen, Lan Li, Libby Lunstrum, Bruce Savage (2021). Advancing resilience science for complex decision-making and governance of energy and water within natural-human-built systems. National Science Foundation Established Program to Stimulate Competitive Research Research Infrastructure Improvement (RII) Track-1 - Phase 2 Developmental Concept Report.
- (3) J. Seth Dustin, R. A. Borrelli (2020). High level examination of Am-241 as an alternative fuel source in radioisotope thermoelectric generators. LA-UR-20-20574.
- (4) R. A. Borrelli, Richard N. Christensen, Brian J. Jaques, Samuel V. Pedersen, Winfred Sowah (2019). Modeling and design of borated aluminum cask for used fuel cooling. APP-002714.
- (5) R. A. Borrelli, Richard N. Christensen, Brian J. Jaques, Samuel V. Pedersen, Winfred Sowah (2019). Modeling and design of borated aluminum cask for used fuel cooling. APP-002714.
- (6) Joonhong Ahn, R. A. Borrelli, Esra Orucoglu, Bret P. van den Akker, Seung Min Woo, Jihae Yoon (2015). Systems assessment for advanced nuclear fuel cycle: Final report for UCBNE-KAERI collaboration. UCBNE-5150.
- (7) Joonhong Ahn, R. A. Borrelli, Jihae Yoon (2012). Systems assessment for advanced nuclear fuel cycle: Fourth year report for UCBNE-KAERI collaboration. UCBNE-5137.
- (8) Siegfried S. Hecker, Edward D. Blandford, Alan S. Hanson, Thomas Isaacs, Niko Milonopoulos, Joonhong Ahn, R. A. Borrelli, Jorshan Choi, Lance K. Kim, Per F. Peterson, Jihae Yoon (2012). Scientific and institutional aspects of advanced systems for spent nuclear fuels in emerging nuclear countries: Final report. UCBNE-5135.
- (9) Joonhong Ahn, R. A. Borrelli, Lance K. Kim, Per F. Peterson, Jihae Yoon, Jorshan Choi (2011). Systems assessment for advanced nuclear fuel cycle: Third year report for UCBNE-KAERI collaboration. UCBNE-5134.
- (10) Joonhong Ahn, R. A. Borrelli, Lance K. Kim, Per F. Peterson, Jasmina L. Vujic, Jihae Yoon, Jorshan Choi, Yongsoo Hwang (2010). Systems assessment for advanced nuclear fuel cycle: Second year

report for UCBNE-KAERI collaboration. UCBNE-5133.

- (11) Joonhong Ahn, Per F. Peterson, R. A. Borrelli, Lance K. Kim, Jihae Yoon (2010). Scientific and institutional aspects of advanced systems for spent nuclear fuels in emerging nuclear countries: Interim report. UCBNE-5130.
- (12) R. A. Borrelli, Olivier Thivent, Joonhong Ahn (2010). Numerical simulation of radionuclide transport in the near field region. UCBNE-5128.
- (13) Joonhong Ahn, Edward D. Blandford, R. A. Borrelli, Eung Ho Kim, Lance K. Kim, Per F. Peterson, Jasmina L. Vujic, Jihae Yoon, Jorshan Choi, Yongsoo Hwang (2009). Systems assessment for advanced nuclear fuel cycle: First year report for UCBNE-KAERI collaboration. UCBNE-5126.
- (14) R. A. Borrelli, Joonhong Ahn (2007). Radionuclide transport coupled with bentonite extrusion in a saturated planar fracture: TRIBEX manual. UCBNE-5111.
- (15) R. A. Borrelli, Joonhong Ahn (2007). Modeling and simulation of bentonite extrusion in a saturated planar fracture. UCBNE-5110.
- (16) R. A. Borrelli, Joonhong Ahn (2007). Modeling and simulation of radionuclide transport in a saturated planar fracture with bentonite extrusion. UCBNE-5109.
- (17) Joonhong Ahn, Paul L. Chambre, Daisuke Kawasaki, R. A. Borrelli (2006). Development of integrated repository performance assessment models. UCBNE-5103.

CONFERENCE PRESENTATIONS

- (1) *Mitchell Meads, R. A. Borrelli, Brittany Brand, Kendra Kaiser, Cassie Koerner (2026). Linking Land-Use Change, Energy–Water Systems, and Ecological Dynamics Through Participatory Scenario Development. Salt Lake City, Utah: Annual Meeting of the Ecological Society of America [*poster*].
- (2) *R. A. Borrelli, Denia Djokić (2026). A Collaborative Based Siting Case Study for Nuclear Siting Tools. Denver, Colorado: Proc., American Nuclear Society Annual Meeting.
- (3) *Haruko M. Wainwright, Shaheen A. Dewji, John McCloy, Brian A. Powell, R. A. Borrelli, Massimiliano Fraton, Jeffrey Wang, Liam Hines, Dauren Sarsenbayev, Craig H. Benson (2026). A Paradigm Shift for High-Level Radioactive Waste Disposal: Radionuclides Intrinsic Properties for Long-term Safety. Denver, Colorado: Proc., American Nuclear Society Annual Meeting.
- (4) Cassie Koerner, *R. A. Borrelli (2025). A Collaborative-Based Siting case study Yucca Mountain siting: The Willrich report predictions. Washington, D. C.: Proc., American Nuclear Society Winter Meeting.
- (5) Eugene Balsmeier, R. A. Borrelli, *Craig L. Porter (2025). Radiological Assessment for Sub-seabed Repositories. Phoenix, Arizona: Proc., Waste Management Symposia.
- (6) R. A. Borrelli, *Kathleen Araújo, Cassie Koerner, Denia Djokić (2024). Consent based siting for Spent Nuclear Fuel – The Common Ground Consortium Focus on Research and Public Conversations. Las Vegas, Nevada: Proc., American Nuclear Society Annual Meeting.
- (7) *Haruko Wainwright, Brian Powell, Megan Hoover, R. A. Borrelli, Carol Eddy-Dilek, Sheldon Landsberger, Rod Ewing (2023). Nuclear Waste Educator’s Workshop: What and How Do We Teach About Nuclear Waste? Indianapolis, Indiana: Proc., American Nuclear Society Annual Meeting.
- (8) *Jonathon Wheelwright, R. A. Borrelli (2023). A Regulatory Approach to Nuclear Reactor Transportation. Washington, D. C.: Proc., American Nuclear Society Annual Meeting.
- (9) *Sam J. Root, Porter Throckmorton, Michael Haney, R. A. Borrelli (2022). Simulated boron shim-ming cyber-attack on pressurized water reactor. Phoenix, Arizona: Proc., American Nuclear Society Winter Meeting.
- (10) *Sam J. Root, Porter Throckmorton, Michael Haney, R. A. Borrelli (2022). Simulated boron shim-ming cyber-attack on pressurized water reactor. [*virtual*]: 2022 3KEYMASTER Consortium Meeting.
- (11) *Michael Haney, Jacob Benjamin, R. A. Borrelli (2021). Cyberweapon Nonproliferation Controls for the Virtual Battlefield - Applying the Nuclear Nonproliferation Regime to an Unseen Enemy.

- Washington, D. C.: Proc., American Nuclear Society Winter Meeting.
- (12) *Samuel Cutler, Allysa Bateman, Brenden Heidrich, R. A. Borrelli, James Simpson, Brian J. Jaques (2020). Modeling shielding designs for the safe operation of neutron generators. Boise, Idaho: Idaho Conference on Undergraduate Research [*poster*].
 - (13) *James Richards, R. A. Borrelli (2020). Development of a tool for evaluating nuclear integrated energy systems profitability for existing reactors. [*virtual*]: American Nuclear Society Virtual Winter Meeting.
 - (14) *R. A. Borrelli (2019). Updates on borated aluminum cask design for onsite used fuel storage. Minneapolis, Minnesota: American Nuclear Society Annual Meeting.
 - (15) *Joseph Christensen, R. A. Borrelli, Charles S. Henkel, Brian Matthews (2019). Minimum accident of concern for uranyl sulfate solutions. Minneapolis, Minnesota: American Nuclear Society Annual Meeting.
 - (16) *Constantinos Koliass, R. A. Borrelli, Daniel Barbara, Angelos Stavrou (2019). Malware detection in critical infrastructures by using the electromagnetic emissions of PLCs. Washington, D. C.: American Nuclear Society Winter Meeting.
 - (17) Trevor MacLean, *Michael Haney, R. A. Borrelli (2019). Cybersecurity modeling of non-critical nuclear power plant instrumentation. Arlington, Virginia: Thirteenth Annual IFIP WG 11.10 International Conference on Critical Infrastructure Protection.
 - (18) *John Peterson, Michael Haney, R. A. Borrelli (2019). An overview of methodologies for cybersecurity vulnerability assessments conducted at nuclear power plants. Washington, D. C.: American Nuclear Society Winter Meeting.
 - (19) *John Peterson, R. A. Borrelli, Dakota Roberson (2019). Informing electrical grid policy with a nuclear power plant simulator. Boise, Idaho: Energy Policy Research Conference [*poster*].
 - (20) *R. A. Borrelli, J. Seth Dustin, Samuel V. Pedersen, Brian J. Jaques (2018). Design of a borated aluminum cask design for onsite used fuel storage. Philadelphia, Pennsylvania: American Nuclear Society Annual Meeting.
 - (21) *R. A. Borrelli, Mark S. Delligatti (2018). Regulatory licensing pathway for a borated aluminum cask design for onsite used fuel storage. Philadelphia, Pennsylvania: American Nuclear Society Annual Meeting.
 - (22) *J. Seth Dustin, R. A. Borrelli (2018). Preliminary performance assessment of americium-241 as fuel in radioisotope thermoelectric generators for deep space exploration. Las Vegas, Nevada: ANS NETS 2018 - Nuclear and Emerging Technologies for Space.
 - (23) *Kathryn D. Huff, Neal Davis, Paul Wilson, R. A. Borrelli, Skutnik Steven E., Anthony Scopatz, Jenny Amos (2018). Open-source curriculum development. Urbana, Illinois: Academy for Excellence in Engineering Education Strategic Instructional Innovation Program at the University of Illinois [*poster*].
 - (24) *Jieun Lee, R. A. Borrelli (2018). Upgrades on high reliability safeguards model for material throughput in fuel fabrication. Philadelphia, Pennsylvania: American Nuclear Society Annual Meeting.
 - (25) R. A. Borrelli, Malachi Tolman, *Jieun Lee (2017). Preliminary analysis of facility design for pyroprocessing safeguardability. San Francisco, California: American Nuclear Society Annual Meeting.
 - (26) *Jieun Lee, R. A. Borrelli (2017). Hazard and operability analysis of a pyroprocessing facility. Washington, D. C.: American Nuclear Society Winter Meeting.
 - (27) *John Peterson, Michael A. Haney, R. A. Borrelli (2017). Cybersecurity vulnerability assessment methodologies for nuclear power plants. San Francisco, California: American Nuclear Society Annual Meeting.
 - (28) *Emma K. Redfoot, Kelley M. Verner, R. A. Borrelli (2017). Applying preliminary hazard analysis and analytic hierarchy process to a nuclear renewable hybrid energy system. Washington, D. C.: American Nuclear Society Winter Meeting.

- (29) *Emma K. Redfoot, R. A. Borrelli (2017). Nuclear fuel cycle simulator as a means to model a nuclear hybrid energy system. San Francisco, California: American Nuclear Society Annual Meeting.
- (30) R. A. Borrelli, *Malachi Tolman (2016). Development of a discrete event simulation model for pyroprocessing safeguardability. Las Vegas, Nevada: American Nuclear Society Winter Meeting.
- (31) *R. A. Borrelli (2013). Summary of modeling studies in the high reliability safeguards approach. Washington, D. C.: American Nuclear Society Winter Meeting.
- (32) *R. A. Borrelli (2013). Applying curium spontaneous fission neutrons in the high reliability safeguards approach for remotely-handled nuclear facilities. Salt Lake City, Utah: Global 2013 - International Nuclear Fuel Cycle Conference.
- (33) *R. A. Borrelli (2012). Applying curium spontaneous fission neutrons in the high reliability safeguards approach for remotely-handled nuclear facilities. Savannah, Georgia: 9th International Conference on Facility Operations-Safeguards Interface.
- (34) *R. A. Borrelli (2011). Role of neutron detection in the high reliability safeguards approach. Albuquerque, New Mexico: International Conf., High-Level Radioactive Waste Management.
- (35) *R. A. Borrelli, Lance Kim, Edward D. Blandford, Yongsoo Hwang, Eung Ho Kim, Per F. Peterson (2010). High reliability safeguards for remote-handled nuclear materials. San Diego, California: 2010 International Congress on Advances in Nuclear Power Plants (ICAPP '2010).
- (36) *R. A. Borrelli, Joonhong Ahn (2010). Parametric studies on confinement of radionuclides in the excavated damaged zone due to bentonite type and temperature change. Nantes, France: Clays in Natural & Engineered Barriers for Radioactive Waste Confinement, 4th International Meeting [poster].
- (37) R. A. Borrelli, *Olivier Thivent, Joonhong Ahn (2009). Effect of temperature change on bentonite extrusion and radionuclide migration in the excavated damaged zone. Luxembourg: Impact of Thermo-Hydro-Mechanical-Chemical Processes on the Safety of Underground Repositories Conference.
- (38) *R. A. Borrelli, Olivier Thivent, Joonhong Ahn (2009). Parametric studies for temperature changes of bentonite extrusion and impacts on radionuclide migration in the excavated damaged zone. Kennewick, Washington: 12th International Conference on the Chemistry and Migration Behaviour of Actinides and Fission Products in the Geosphere [poster].
- (39) *R. A. Borrelli, Joonhong Ahn (2008). Impact of bentonite extrusion on radionuclide migration in a saturated planar fracture. Osaka, Japan: 2008 Annual Meeting of the Atomic Energy Society of Japan.
- (40) *R. A. Borrelli, Joonhong Ahn (2007). Impact of bentonite extrusion on radionuclide in a saturated planar fracture. Lille, France: Clays in Natural & Engineered Barriers for Radioactive Waste Confinement, 3rd International Meeting.
- (41) *R. A. Borrelli, Joonhong Ahn (2007). Radionuclide transport in a water-saturated planar fracture with bentonite extrusion. San Francisco, California: Transport Behavior in Heterogeneous Polymeric Materials and Composites, Mater. Res. Soc. Symp. [poster].
- (42) *R. A. Borrelli, Joonhong Ahn (2006). Bentonite extrusion in planar fracture based on net water flow. Las Vegas, Nevada: International Conf., High-Level Radioactive Waste Management.

OTHER PRESENTATIONS

Invited talks

International Atomic Energy Agency

Knowledge Management Assist Visits for Education and Training Providers

2021 virtual

University of Idaho · Idaho Falls Center for Higher Education

Strategies and success for ethical research

2018–19; 2022

Vandal Advantage Graduate Student Orientation

2017–19

P3/TRIO Upward Bound STEM Day Presentation

2015

Stanford University
Pre-Collegiate Summer Institutes 2014

Seminars

University of Tennessee–Knoxville Nuclear Engineering Department 2019

Lightning Talks

Future of Nuclear Waste Education – Massachusetts Institute of Technology 2022
Idaho National Laboratory Nuclear Science and Technology Collaborative Planning Meeting 2018
Idaho National Laboratory National and Homeland Security Collaborative Planning Meeting 2018
Idaho National Laboratory Energy & Environment Collaborative Planning Meeting 2017

STUDENTS ADVISED

University of Idaho *Affiliation is of where majority degree research was conducted*
Major Professor

Current

Dylan Ohrt	Idaho National Laboratory	Nuclear materials safeguards	Ph.D.
J. Seth Dustin	Oklo	Waste processing	Ph.D.
Nathan Manwaring	Idaho National Laboratory	Criticality safety	Ph.D.
Trevin Lasley	Naval Reactors Facility	Criticality safety	Ph.D.
David Haar	Self-employed	Decontamination & Decommissioning	Ph.D.
Evan Able	Curtiss-Wright Nuclear (Scientech)	Advanced reactor neutronics	MS
Kevin Haar	Waste Isolation Pilot Plant		MS

Graduates

Olin Calvin	University of Idaho	Depletion chain simplification with pseudo-nuclides to model decay effects	Ph.D.	2023
Joseph Christensen	TerraPower	Advancements in the Evaluation of Heterogeneity for Nuclear Criticality Safety in High-Assay Low-Enriched Uranium Systems	Ph.D.	2023
Teyen Widdicombe	University of Idaho	Investigation of Interactions Between Radiations from Dragonfly's MMRTG and Titan's Environment	Ph.D.	2022
Kelley M. Verner	University of Idaho	Irradiation induced phase change in low enriched uranium-molybdenum fuel as it relates to microstructure	Ph.D.	2021
Ethan Bauer	Idaho National Laboratory	Detector design	MS	2026
J. Seth Dustin	University of Idaho	High level examination of Am-241 as an alternative Fuel Source in radioisotope thermoelectric generators	MS	2020
John Peterson	University of Idaho	Molten salt reactor neutronics design	MS	2019

Jieun Lee	University of Idaho	Risk-informed safeguards of pyroprocessing for advanced nuclear fuel concepts	MS	2018
Emma Redfoot	University of Idaho	Allocating heat and electricity in a nuclear renewable hybrid energy system coupled with a water purification system	MS	2018
Malachi Tolman	University of Idaho	INL BISON code documentation	MS	2017
Jonathon Wheelwright	Idaho National Laboratory	Microreactor transport	MEng	2023
Trevor MacLean	Idaho National Laboratory	Cybersecurity modeling of non-critical nuclear power plant instrumentation	MEng	2018

**University of Idaho
Committee Member**

Graduates

Jonathan Tacke	University of Idaho	Frequency Regulation by Way of a Variable Moment of Inertia	Ph.D.	2024
Ryan C. Hruska	Idaho National Laboratory	A Functional All-Hazard Approach to Critical Infrastructure Dependency Analysis	Ph.D.	2023
David Kamerman	Idaho National Laboratory	On the Role of Bulk Hydrides and Hydride Rims in Causing Low Temperature Ruptures of Zircaloy-4 Cladding Tubes in Transient Reactor Tests	Ph.D.	2023
James Richards	Ultra Safe Nuclear Corp.	Techno-Economic Analysis of Nuclear Integrated Energy Systems for Water Desalination and Hydrogen Production	Ph.D.	2023
John Carter	Idaho National Laboratory	Core design of molten salt nuclear battery	Ph.D.	2022
Jacob Benjamin	Dragos, Inc.	Bounding cyber in design basis threat	Ph.D.	2020
Hunter Squires	University of Idaho	Industrial Control Systems Security Education	MS	2026
Gabriel Lewis	University of Idaho	Redox measurement and Corrosion testing in molten FLiNaK salts	MS	2025
Sam J. Root	University of Idaho	Dynamic System Modeling & PID Controller Design for a Molten Salt Microreactor	MS	2024
Stefan Abbot	University of Idaho	Electrochemical Analysis of Molybdenum, TZM, and Molybdenum-Lanthanum ODS in Aqueous Solution	MS	2024
Kendall Bean	University of Idaho	Using static VAR Compensators to Simultaneously Regulate Power System Voltage and Frequency	MS	2022
Trevin Lasley	University of Idaho	Criticality safety on the molten salt nuclear battery	MS	2021
Joshua Young	University of Idaho	Analysis of a Dump Heat Exchanger For The Versatile Test Reactor's Secondary Loop	MS	2021
Jonathan Tacke	University of Idaho	Design of an automatic voltage regulator with limited plant information	MS	2020

John Bell	University of Idaho	Hierarchical inference and spoofing alarm in HVDC control systems	MS	2020
Winfred Sowah	University of Idaho	Thermal behavior of cold plated storage cask for used light water reactor nuclear fuels	MS	2019
D. Devin Imholte	Idaho National Laboratory	Conceptual design of the Advanced Test Reactor non-destructive examination system	MS	2019
John Biersdorf	University of Idaho	Precipitation changes to Idaho National Laboratory over time	MEng	2018

Idaho State University

Committee Member

Graduates

Pedro Mena	Idaho State University	Reactor transient classification using machine learning	Ph.D.	2012
Pedro Mena	Idaho State University	Auto Machine Learning Applications for Nuclear Reactors: Transient Identification, Model Redundancy and Security	MS	2019

UNIVERSITY SERVICE

University of Idaho

Chair

Faculty Affairs Committee	2025–26
Promotion & Tenure – Nuclear Engineering & Industrial Management	2023
Third Year Promotion & Tenure – Nuclear Engineering & Industrial Management	2021

Member

University Level Promotions and Tenure Silver Committee	2026
University Level Promotions and Tenure Gold Committee	2025
Center for Excellence in Teaching and Learning Accessibility Advisory Committee	2025–6
Faculty Compensation Committee	2025–6
Promotion – Chemical & Biological Engineering	2025
Strategic Planning Focus Group	2025
Scientific Misconduct Committee	2025–26
Faculty Senate – College of Engineering	2024–27
Faculty Affairs Committee	2024–26
Faculty Appeals Hearing Board Committee	2024–25
Scientific Misconduct Committee Alternate	2024–25
Promotion & Tenure, College of Engineering	2020–23
Nuclear Regulatory Commission Student Fellowship Oversight Committee	
Nuclear Engineering & Industrial Management Bylaws Committee	
Graduate Faculty	
Nuclear Engineering & Industrial Management Admissions Committee	
Commencement Speaker Committee – Idaho Falls Center for Higher Education	

Organizer

American Nuclear Society Graduate School Fair	2020 – 22
NNSA Fellowship Information Symposia for Idaho Universities	2017 –

Faculty Advisor – American Nuclear Society Student Section	2015–
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Faculty Search Committees

Chair

Nuclear Engineering & Industrial Management <i>Member</i>	2024
Grants & Contracts Specialist II	2024
Nuclear Engineering & Industrial Management	2023
Nuclear Engineering	2022
Mechanical/Nuclear Engineering	2017

Idaho State University

Faculty Search Committees

Nuclear Engineering	2018
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PROFESSIONAL & SCHOLARLY ORGANIZATIONS

National

American Nuclear Society

National Program Screening Subcommittee	2022–
Local Sections Committee	
Member-at-Large	2025–28
Fuel Cycle & Waste Management Division	2015–
Waste Management Certificate Committee	2025–
Executive Committee	2018–21; 2026–29
Student Support Subcommittee	2025
Nonproliferation Policy Division	2015–
Executive Committee	2019–22
Student Sections Committee	2015–
Executive Committee	2018–21; 2024–27

<i>Tau Beta Pi Engineering Society</i>	1996
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Regional

Idaho Section of the American Nuclear Society

Treasurer	2022–
Board of Directors	2018; 2020
Nuclear Science Week Planning	2017
Community Service	2015–

PROFESSIONAL & SCHOLARLY ACTIVITIES

Special Editor

Nuclear Technology

Participatory Decision-making for the Nuclear Fuel Cycle

In recent years, the United States shifted toward a more participatory decision-making model for siting facilities in the nuclear fuel cycle. This recognizes revised thinking, championed by the Blue Ribbon Commission, that more enduring nuclear waste management hinges on factors that extend beyond geology and engineering. Such factors include collaboration with community and Tribal partners, procedural fairness, and inter/cross-jurisdictional alignment, among other considerations. This special issue seeks to explore the multidisciplinary challenges of managing the nuclear fuel cycle in the context of participatory decision-making.

Conference Committees

American Nuclear Society

Technical Program, International High Level Radioactive Waste Management, Washington, D.C.	2025
Technical Program, Advances in Nuclear Nonproliferation Technology and Policy, Orlando, Florida	2016; 2018
Co-Chair	2018

Technical Session Organizer

American Nuclear Society

Cybersecurity for Nuclear Installations, Washington, D. C. 2019; 2021
co-Organizer with Prof. Jamie B. Coble, University of Tennessee–Knoxville

Technical Session Chair

American Nuclear Society

Program Development, Strategy, and Policy: II, International High Level Radioactive Waste Management, Washington, D. C. 2025

Cybersecurity for Nuclear Installations, Anaheim, California 2022

Technology and Policy Advancements in Nuclear Nonproliferation Anaheim, California 2022

Cybersecurity for Nuclear Installations, Washington, D. C. 2019; 2021

Spent Fuel Storage and Transportation, Washington, D. C. 2019

Data Synthesis for Pyroprocessing Safeguards, Advances in Nuclear Nonproliferation Technology and Policy, Orlando Florida 2018

Prof. Joonhong Ahn Memorial Session, International High-Level Radioactive Waste Management, Charlotte, North Carolina 2017

Nonproliferation Policy, Concepts, and Approaches: Treaty verification regimes, State-level Concepts, and Fuel Cycle Analysis, Advances in Nuclear Nonproliferation Technology and Policy, Santa Fe, New Mexico 2016

Security, Safeguards, and Non-proliferation, International High-Level Radioactive Waste Management, Albuquerque, New Mexico 2011

Used Fuel Recycling Technologies–I, International Congress on Advances in Nuclear Power Plants, San Diego, California 2010

Engineered Systems and Transport Processes, International High-Level Radioactive Waste Management, Las Vegas, Nevada 2008

Boise State University Energy Policy Institute

Diverse Approaches to Addressing Decarbonization, Energy Policy 2021 virtual

University of Tokyo

Challenges in Nuclear Waste Disposal: Sociological Aspects and Technical Approaches, Global Center of Excellence 2007

Panels

Moderator

American Nuclear Society

Future of Cybersecurity in Nuclear Installations 2021

Panelist

American Nuclear Society

Future of Cybersecurity in Nuclear Installations 2021

Taking Care of You: Student-Mentor Relationships 2021

A great mentoring situation can pave the way for success out of undergraduate or graduate school. Likewise, a poor or unhealthy mentoring relationship can cause lasting negative consequences on the students. Many times these interactions go undiscussed, and one goal of this workshop is to empower students to talk about and take charge of unhealthy mentoring relationships.

University of Idaho

Idaho Open Education Week Think Open Fellowship 2021

Think Open Fellows an incubator program that awards 6 fellowships each year to support faculty and graduate students in transitioning a course from a standard text to open or extremely low cost course materials.

Boise State University Energy Policy Institute Power Talks

Economic Opportunities and Challenges for Idaho with Low Carbon Energy 2022

Workshops

Idaho Community-Engaged Resilience for Energy-Water Systems

Machine Learning, Computational Modeling, Risk Assessment 2024

Presenter

[virtual]

The workshop is designed to provide an introductory overview of machine learning, including what machine learning is, different types of learning, and machine learning in practice. The workshop will provide machine

learning examples used in water and energy research, followed by discussion. Additionally, the workshop will feature discussions on energy modeling with a focus on timescales, and an exploration of risk, including its dynamic nature and examples from the literature. The objective of the workshops are to provide an opportunity to enhance participant understanding of machine learning, fostering collaboration and knowledge exchange within the I-CREWS community.

Future of Nuclear Waste Education

Clemson University

2023

Presenter

Understanding and managing nuclear waste requires a broad knowledge from nuclear engineering to civil and environmental engineering. In parallel, broader system-level analyses as well as societal/regulatory contexts are necessary to inform energy policies from the nuclear waste and environmental perspectives. Energy system or nuclear fuel cycle analysis quantifies waste generation per unit energy output, and their economical and environmental impacts throughout the life cycles of nuclear energy. Historical and regulatory contexts are also important to understanding nuclear waste in the contexts of general hazardous waste management. A challenge is posed by the extreme interdisciplinary nature of the subject of nuclear waste management— which makes it difficult to cover all the important topics in one, single institution. Such a knowledge gap is a major bottleneck when transformational changes are needed to prepare for new types of wastes from advanced reactors as well as to develop new waste processing technologies such as pyroprocessing. Although each individual institution can have its strengths and characteristics, it is critical to create community efforts such that the next generation workforce – interacting with the public and policy makers – has current and comprehensive knowledge about nuclear waste and its environmental impacts. This workshop will develop the blueprint of educating next generation engineers/scientists in nuclear waste as well as broader nuclear and environmental engineers. Although one student may not need to have all the knowledge, there is a minimum set of knowledge that one should have as a future scientist or engineer involved in nuclear waste and nuclear energy.

Nuclear Advocacy and Communications Training Generation Atomic

Idaho Falls

2018

Facilitator

Opposition groups claim nuclear power plants are unsafe. Recently, the U.S. nuclear power industry has been characterized as too expensive and dangerous when compared to other energy sources. As members of the nuclear community, we know that the success of nuclear energy has never been more important to ensuring a positive future for the world – but what can we do to make a difference? This workshop will leave participants energized to tell today’s nuclear power story and be well-equipped with the tools to do so. Convincing others about the benefits of nuclear involves more than just laying out the facts. Thoughtful and personal storytelling bridges gaps when speaking those who are unfamiliar with the technology by explaining the personal and moral reasons that we work in this field. Telling our personal stories and motivations for working in nuclear creates common ground from which we can better explain nuclear’s benefits: whether it’s as a mother talking to a father, a surfer talking to a skier, or a cook talking to a conservationist, the human stories that nuclear makes possible are our strongest tools when speaking to the public. The most effective nuclear advocacy takes place at the interpersonal level when we strike up conversations with peers and even better, strangers. Because you can never know who it might be sitting across from you at that dinner party or next to you on the airplane, it’s important to practice having open, considerate conversations with people of all backgrounds.

Modeling, Experimentation, Validation Summer School

Idaho National Laboratory

2017

Mentor

The MeV Summer School provides enhanced training for engineers and applied scientists involved in design, licensing, and operation of current and advanced nuclear reactor systems through a multi-faceted learning approach of lectures, tours, and hands-on activities. The school is being organized through the cooperation of national laboratories, industry, government agencies, and universities that share the goal of building a strong workforce to support global nuclear expansion. The faculty will be drawn from the top experts in academia, industry, and government. The general organization and conduct of the school will be overseen by an international board of senior

experts. A local secretariat will provide technical, logistical and administrative support to students and faculty. It is the aim of the school to foster the development of a next-generation network of scientists and engineers capable of advancing nuclear energy in the 21st century through integrated modeling and experimentation.

Symposia

Organizer

Critical Materials and Energy Systems Innovation Center

Nuclear cybersecurity research initiatives

2017

Nuclear cybersecurity research focus areas

2016

Stanford University

Center for International Security and Cooperation

Technical implications of nuclear energy system options

2011

PEER REVIEWER

American Nuclear Society

Advances in Nuclear Nonproliferation Technology and Policy

Annual & Winter Meetings

Fuel Cycle and Waste Management Division John Randall Scholarship

Chair

2022 – 23

Instrumentation, Control & Human-Machine Interface Technologies

International High-Level Radioactive Waste Management Conference

Student Conferences

Journals

Advances in Engineering Software

Annals of Nuclear Energy

Energy Science & Engineering

IEEE Transactions on Nuclear Science

International Journal of Nuclear Energy

Nuclear Engineering and Design

Nuclear Engineering and Technology

Progress in Nuclear Energy

Government

Nuclear Science User Facilities

DOE Advanced Research Projects Agency-Energy Concept Papers

DOE Advanced Research Projects Agency-Energy Review Panel

DOE Office of Nuclear Energy, Consolidated Innovative Nuclear Research

DOE SBIR/STTR Phase I Release 2

DOE SBIR/STTR Phase II Release 1

DOE SBIR/STTR Phase II Release 2

Other

ORAU Ralph E Powe Junior Faculty Enhancement Awards

Louisiana Board of Regents Support Fund R & D Grants Programs

Clays in Natural & Engineered Barriers for Radioactive Waste Confinement

Bonneville Power Administration, Office of Technology Innovation

John Wiley & Sons, Inc.

Khalifa University of Science and Technology

TEACHING EXPERIENCE

University of Idaho · Idaho Falls Center for Higher Education

Nuclear Engineering & Industrial Management

Used Fuel Management & Nuclear Power Plant Decommissioning Certificate *Offered variably as needed*

NE514: Nuclear Safety

The focus of this course is on the approach to nuclear safety for the commercial nuclear industry. Historical events serve as a basis for learning to investigate how nuclear energy policy has affected approaches to safety developed in the United States. Additionally, how safety approaches affect the nuclear industry will be covered. Several different classes of nuclear facilities will be studied. Relevant current events will be highlighted.

NE516: Nuclear Rules & Regulations

The focus of this course is on the rules and regulations that govern the commercial nuclear industry. Historical events serve as a basis for learning to investigate how nuclear energy policy has developed in the United States. This includes not only current regulations but important development that led to major policy changes. Additionally, how current regulations affect the nuclear industry will be covered. Facilities studied will include research, commercial, and government. Relevant current events will be highlighted.

NE527: Nuclear Materials, Transportation, Storage

in development

The focus of this course is on the scope and sequencing of activities necessary to comply with the rules and regulations that govern storage, transport, and disposal of nuclear materials. The course is mainly focused on spent nuclear fuel management, Department of Energy high-level waste management, and NRC regulated activities. The course is based on United States nuclear facilities and nuclear power plants.

NE587: Nuclear Facility Decommissioning

The focus of this course is on the scope and sequencing of activities necessary to comply with the rules and regulations that govern decommissioning of nuclear facilities. The course is mainly focused on the NRC regulated activities, as these are both stricter and more regimented when compared to the DOE/DOD regulated activities. However, most of the rules apply to both. The course is based on D&D experience in the US, both for facilities which have successfully completed the process and for facilities currently in the process.

University of Idaho · Idaho Falls Center for Higher Education
Nuclear Engineering & Industrial Management

Regularly scheduled

NE587: Nuclear Facility Decommissioning

Fall 2025

supervising instructor

The focus of this course is on the scope and sequencing of activities necessary to comply with the rules and regulations that govern decommissioning of nuclear facilities. The course is mainly focused on the NRC regulated activities, as these are both stricter and more regimented when compared to the DOE/DOD regulated activities. However, most of the rules apply to both. The course is based on D&D experience in the US, both for facilities which have successfully completed the process and for facilities currently in the process.

NE585: Nuclear Fuel Cycle Analysis

Fall 2023–; 2017

This course presents the nuclear fuel cycle can as an holistic system with components related in many complex ways. This course focuses on systems analysis of components that comprise the nuclear fuel cycle to understand the contemporary challenges facing nuclear energy. Topics include reactor design, critical size, reactor statics and dynamics, advanced reactor design, and back-end management; siting, fuel management, interim storage, repository design. Students will also gain facility with MCNP.

NE529: Risk Assessment

Spring 2021–; 2016–19

This course is designed to provide students with an understanding of how to perform a comprehensive risk assessment applicable to a wide variety of engineering problems in many different disciplines. The course will focus on failure mode and effect analysis, fault tree analysis, probabilistic risk analysis, and human reliability analysis. The course will also cover fundamental probability and statistics content.

NE544: Reactor Analysis – Statics and Kinetics

*Spring 2022 –
supervising instructor*

The purpose of this course is to study nuclear theory in the context of nuclear reactor engineering; concepts relating to the design and operation of nuclear reactors will be discussed. Content includes development of mathematical expressions describing the relevant nuclear processes as well as their physical implications. This course will involve the application of several common analytical tools used for the design and evaluation of nuclear systems.

NE450: Principles of Nuclear Engineering

Fall 2015–22

In this course, an overview of fundamental nuclear engineering principles and how these are practically applied to contemporary, nuclear engineering problems will be presented. The topics covered in this course include: nuclear physics and reactions, materials science, radiation protection, energy production, fuel cycle analysis, advanced reactor design, fusion, nonproliferation, back-end management, and risk assessment and safety. Throughout the course, the ethical considerations with regards to engineering problems within these fields will also be discussed.

NE502: Python MCNP Development for Molten Fuel Salt Handling

Spring 2025

In this course, a series of Python Scripts are developed to construct MCNP input decks that will be applied to establish criticality safety fuel handling controls for molten fuel salt.

NE502: Nuclear Power Plant Decommissioning Education

Fall 2023

As variable energy sources increasingly penetrate the United States energy market, the economics of nuclear energy has weakened. This has led to planned plant closures. Under NRC regulations, these plants must be decommissioned; i.e., safe removal from service and reduction of radioactivity to permissible levels in order to release the property. These activities will require specialized training. The market for decommissioning in North America was valued at \$1.92B in 2019, and it is projected to increase to \$3.35B in 2027. A significant, skilled, and technically proficient workforce will be needed to complete these activities safely and within regulations. Currently, there are no programs in the US specifically focused on decommissioning workforce training. This course will develop a nuclear power plant decommissioning asynchronous, online graduate course as part of the University of Idaho Nuclear Power Plant Decommissioning Graduate Certificate.

NE502: Computation and modeling of nuclear systems

Fall 2020

Using published nuclear computational modeling references, this course will examine prior efforts to simplify the neutron transport and depletion calculations for advanced reactor designs, such as high-temperature pebble-bed reactors. These include efforts to model heterogeneous stochastic media, such as TRISO fuel particles dispersed in a graphite matrix. Deterministic neutron transport in stochastic media has resulted in several new methods developed in order to address the challenges associated with modeling them, however, comparison of the proposed and implemented methods has been largely absent. Other challenges facing the modeling of pebble-bed reactors include the depletion calculations for individual pebbles. Because of the flow of pebbles through the core, and the recirculation of said pebbles in most modern designs, the depletion of individual pebbles can affect the criticality of the system as well as the temperature profile of individual pebbles, which was demonstrated to have significant safety implications based on the operation of the German AVR pebble-bed system. To these ends, this course will identify relevant literature regarding these topics and the feasibility of implementing these methods in modern nuclear modeling tools, such as the Griffin tool in development at Idaho National Laboratory.

NE502: Heterogeneity comparisons of intermediate enrichment uranium in critical systems

Fall 2020

The ICSBEP handbook contains descriptions and evaluations of critical experiments conducted in facilities throughout the world. The reach of the handbook stretches to the beginning of the history of nuclear engineering. As part of that history, a number of critical experiments have been conducted using arrays of fissile material in the intermediate enrichment range, which is defined for the purposes of this course as ten to forty percent (10%-40%). In those evaluations, a variety of methods have been discussed and described which convert a system of known or unknown degree of heterogeneity into a simplified homogeneous model, which is a traditional method for examining the critical characteristics of multiplying systems. The specific methodology for this conversion varies between evaluations and is explained in some cases, where other cases are less descriptive. It is of interest to the field of nuclear criticality safety to improve the understanding of the effect of heterogeneity on the critical characteristics of multiplying fissile systems, particularly in the range of intermediate enrichment. It is of further interest to the field that a methodology be established which can be used to evaluate these types of systems for practical nuclear

criticality safety applications, especially the development of nuclear criticality safety limits.

NE502: Computation of spherical critical volume for nuclear criticality safety applications

Fall 2020

The nuclear safety guide, Critical Dimensions of Systems Containing U-235, Pu-239, and U-233 contains a graph of minimum critical volumes as a function of U-235 enrichment in homogeneous and heterogeneous hydrogen-moderated systems. This graph includes a wide extrapolation between the data for ten percent enrichment and ninety-three percent enrichment. A similar graph appears in Anomalies of Nuclear Criticality without the extrapolation. This graph shows a clear transition at approximately thirty-five percent enrichment between two different types of systems. It is clear that there is a difference between the extrapolated minimum spherical critical volumes and the calculated critical volumes. As part of the development of nuclear criticality safety methodology for intermediate-enrichment uranium systems, it is necessary to more fully evaluate and explain the nature of the transition point. This work will attempt to re-create and expand on the data used to produce this graph and more fully evaluate the nature of the minimum critical volume for intermediate-enrichment systems. The secondary objective will be the proposal of nuclear criticality safety rules which could be applied to these data to ensure that fissionable material operations in this range of enrichment can be conducted safely.

NE502: Subcritical multiplication

Fall 2020

Using published fundamental nuclear data for thermal neutrons, this course will examine the relationship between processes of fission and capture and develop models for reactivity worth as a function of thermal fission and capture cross-sections and local thermal flux. This course will include evaluation of tests in zero-power reactors. The Advanced Test Reactor (ATR) physics analyses rely heavily on measurement results from the Advanced Test Reactor Critical Facility (ATRC). These results are easily delivered in terms of reactivity worth, but the results are almost never directly applicable to ATR due to differences in thermal neutron flux. Reactivity worth is not a fundamental property of a material and indeed is not a characteristic property of a given piece of reactor hardware if the hardware is to be irradiated in a new neutron environment. However, translating reactivity worth of a piece of hardware in ATRC to a pair of macroscopic cross-sections essentially characterizes the hardware with constants that will be applicable in any subsequent location in ATR or ATRC. Therefore, reactivity worth in a new irradiation position can be predicted by calculation.

NE502: Historical examinations of heterogeneity in nuclear criticality safety

Spring 2020

Using published nuclear criticality safety references, this course will conduct an examination of the effect of heterogeneity with respect to nuclear criticality safety. As part of the history of nuclear criticality safety, a number of critical experiments have been conducted using arrays of fissile material in an attempt to quantify the effect of heterogeneity in the determination of safe handling limits. In these attempts, a number of low-enriched experiments were examined and converted in their critical dimensions to establish a useful baseline from which other limits could be extrapolated. It is of interest to the field of nuclear criticality safety to improve the understanding of the effect of heterogeneity on the critical characteristics of multiplying fissile systems, particularly in the range of intermediate enrichment. It is of further interest to the field that a methodology be established which can be used to evaluate these types of systems for practical nuclear criticality safety applications, especially the development of nuclear criticality safety limits. To that end, this course will identify relevant data in the literature used to generate historical experiments and apply MCNP to model these experiments in order to understand and evaluate assumptions and restrictions raised in the experiments.

NE502: Heterogeneity comparisons of intermediate enrichment uranium in critical systems

Spring 2020

Using the International Criticality Safety Benchmark Evaluation Project (ICSBEP) handbook of evaluated critical experiments, conduct an examination of the effect of heterogeneity in intermediate-enrichment uranium systems. The ICSBEP handbook contains descriptions and evaluations of critical experiments conducted in facilities throughout the world. The reach of the handbook stretches to the beginning of the history of nuclear engineering. In those evaluations, a variety of methods have been discussed and described which convert a system of known or unknown degree of heterogeneity into a simplified homogeneous model, which is a traditional method for examining the critical characteristics of multiplying systems. The specific methodology for this conversion varies between evaluations and is explained in some cases, where other cases are less descriptive. It is of interest to the field of nuclear criticality safety to improve the understanding of the effect of heterogeneity on the critical characteristics of multiplying fissile systems, particularly in the range of intermediate enrichment. It is of further interest

to the field that a methodology be established which can be used to evaluate these types of systems for practical nuclear criticality safety applications, especially the development of nuclear criticality safety limits. This course will identify and evaluate relevant benchmark experiments for heterogeneity effects using MCNP to establish new baseline models from the benchmark handbook.

NE502: Nuclear integrated energy systems

Spring 2020

Nuclear renewable hybrid energy systems enable a nuclear reactor to load follow with a renewable energy source. These must be designed to distribute energy dynamically by supplying electricity to the grid while using either thermal or electrical energy for industrial applications. This takes advantage of the flexible distribution of electricity or heat to maximize profit. The industrial process serves as a load sink for the excess heat or electricity produced by the nuclear reactor. Economic challenges to the current United States nuclear light water reactor (LWR) fleet have led to early plant closures. While LWRs primarily deliver baseload electricity, there is no reason why nuclear energy produced by these reactors cannot be used to provide energy to a range of industrial applications. This directed study course will identify feasible systems, products, and commodities that could be produced by existing nuclear plants. As part of this, cost and potential profitability will be analyzed within the context of market structures and grid reliability.

NE527: Nuclear material storage, transport, disposal

Spring 2020

There is a wide range of nuclear materials that are stored, transported and disposed of each day. The materials include medical radioisotopes, new fuel pellets, used fuel, and industrial radioisotopes. This course will cover the regulations that govern nuclear material storage, transportation and disposal, as well as the engineering requirements and practical aspects of handling these materials.

NE535: Nuclear Criticality Safety I

Spring 2022; 2020

This course applies uses the foundation of applied nuclear physics to develop and explain the international and domestic rules and practices that are used to prevent inadvertent criticality in fuel cycle applications such as used fuel storage and processing.

Diablo Valley College
Architecture and Engineering
Adjunct Professor

ENG110: Introduction to Engineering

2013–15

This course introduces students to fundamental engineering principles. Students learn how these are applied to contemporary engineering problems through laboratory exercises, homework assignments, design projects, interviews with professional engineers, and field trips to engineering companies. Topics include: materials science, risk assessment and safety, critical problem-solving, engineering analysis, engineering design processes, project development, engineering software, and presentation tools. The role of the engineer in society, professionalism, and engineering ethics are major themes. The emphasis is on creative problem-solving, teamwork, and effective communication, both in presentation and writing.

University of California –Berkeley
Nuclear Engineering
Instructor

NE375: Teaching Techniques in Nuclear Engineering

2006; 2010–11

This course acquaints Graduate Student Instructors (GSIs) with teaching techniques for courses in the Department of Nuclear Engineering. The GSI will have several duties far beyond grading assignments and/or examinations: conducting discussion sessions, review lectures, or laboratory experiments. The GSI, therefore, needs to develop the appropriate tools to use when facing these pedagogical challenges. Three students from the 2010 course received the Outstanding Graduate Student Instructor Award given by the UC-Berkeley Graduate Student Instructor Teaching & Resource Center.

Graduate Student Instructor

E124: Ethics and the Impact of Technology on Society

2004–06

Conducted multiple discussion sections on a weekly basis and review lectures
 Supervised research projects based on current, ethical and scientific issues
 Assessed individual student presentations, projects, and overall course grading

IDS110: Introduction to Computing 2004

Conducted multiple laboratory sessions on a weekly basis
 Supervised undergraduate research projects focused on web based education
 Assessed laboratory assignments and project grading

NE92: Issues in Nuclear Science and Engineering 2000; 2002

Conducted multiple laboratory sessions on a weekly basis
 Supervised undergraduate research projects focused on web based education
 Assessed laboratory assignments and project grading

NE275: Principles and Methods of Risk Analysis 2001

This graduate course requires a deeper understanding of the subject matter, due to the student body. The course was one of three in the curriculum with the highest credit load. The main responsibility in this was to advise and grade semester projects and presentations based on risk assessments of engineering systems.

Reader

NE150: Introduction to Nuclear Reactor Theory 2003

NE104: Radiation Detection and Nuclear Instrumentation Laboratory 2002

NE107: Introduction to Imaging 2001

NE120: Nuclear Materials 2000

Supervised laboratory sessions
 Assessed examinations, homework assignments, laboratory reports, final grades
 Conducted review lectures

University of Tokyo

Nuclear Engineering & Management

Part Time Lecturer

Technical English for Scientists 2007–08

This course provided the opportunity for non-native English speaking students to develop technical communication skills; i.e., presenting scientific and technical material to an informed audience at an international conference. In this course, the ‘assertion evidence design’ concept for technical presentation of scientific topics was applied to student research interests. Transmutable skills focused on the professional communication of scientific research in various public speaking formats and a comfortable familiarity with the English language to establish a stronger foundation for technical writing.

TEACHING MATERIALS DEVELOPED

Open Educational Resources

Open source educational materials were compiled in order to produce an online textbook as a supplement to existing commercial textbooks. An online textbook allows flexibility to augment course content without requiring multiple textbooks. It also allows the educational content of a course to be more closely aligned with the desired learning outcomes. Two online texts have been developed – [Principles of Nuclear Engineering](#) and [Risk Assessment](#).

OUTREACH

University of Idaho

University of Idaho, Utah State University, Boise State University American Nuclear Society Student Social 2022

American Nuclear Society Diversity and Inclusion Committee Sponsorship 2020–

Utah State University Graduate School Fair 2019

Montana Tech Career Fair 2018–22

University of Idaho Moscow Campus Recruiting 2017–

Live After 5 Idaho Falls	2017
United States Department of Energy consent based siting meeting – Boise	2016
PHYSOR University of Idaho Idaho Sponsorship	2016
Boise State University Nuclear Research Summit	2016
University of Idaho & Brigham Young University American Nuclear Society Student Social	2016; 2018
Brigham Young University–Idaho Career Fair	2015

Regional

Idaho Section of the American Nuclear Society

Highway Cleanup 2016–

Garbage cleanup of Miles 122-124 on Interstate 15 biannually

Smoke Detector Donation Program

Since 2008, we have worked with more than 75 fire departments to donate more than 6200 smoke detectors to Idaho residents. From 2016, we donated over 2300 smoke detectors to Arimo, Ashton, Bancroft, Bear Lake, Clear Creek, Declo, Downey, Driggs, Grace, Hamer, Lava Hot Springs, Roberts, Shelley, and Soda Springs, across southeastern Idaho; Centerville, Clear Creek, Gem County, Horseshoe Bend, Idaho City, Lowman, Placerville and Valley of the Pines in Boise County; and north to the panhandle in Coeur D’Alene, Mullan, Shoshone, North Side, and West Pend, as well as nonprofits such as Habitat for Humanity and Club Inc. In 2022, we were pleased to be able to expand into Western Wyoming to Teton County.

College of Eastern Idaho Machine Tool Technology Advisory Board 2018

PROFESSIONAL DEVELOPMENT

(Re)Connecting with the 6Rs

A Workshop Series on Ethical Native-Engaged Research Respect, Relationality, Responsibility, Representation, Relevance, and Reciprocity

2026

This Reciprocal Research Workshop series, funded by the NSF EPSCoR I-CREWS grant, employs the 6 Rs (Respect, Relationality, Responsibility, Representation, Relevance, Reciprocity) framework for exploring the dynamics, logistics, and practical considerations of conducting or engaging in collaborative research with, in, and by Native Nations.

Future of Nuclear Waste Education Workshop

Vanderbilt University 2024

Clemson University 2023

Massachusetts Institute of Technology 2024

Understanding and managing nuclear waste requires a broad knowledge from nuclear engineering to civil and environmental engineering. In parallel, broader system-level analyses as well as societal/regulatory contexts are necessary to inform energy policies from the nuclear waste and environmental perspectives. Energy system or nuclear fuel cycle analysis quantifies waste generation per unit energy output, and their economical and environmental impacts throughout the life cycles of nuclear energy. Historical and regulatory contexts are also important to understanding nuclear waste in the contexts of general hazardous waste management. A challenge is posed by the extreme interdisciplinary nature of the subject of nuclear waste management— which makes it difficult to cover all the important topics in one, single institution. Such a knowledge gap is a major bottleneck when transformational changes are needed to prepare for new types of wastes from advanced reactors as well as to develop new waste processing technologies such as pyroprocessing. Although each individual institution can have its strengths and characteristics, it is critical to create community efforts such that the next generation workforce – interacting with the public and policy makers – has current and comprehensive knowledge about nuclear waste and its environmental impacts. This workshop will develop the blueprint of educating next generation engineers/scientists in nuclear waste as well as broader nuclear and environmental engineers. Although one student may not need to have all the knowledge, there is a minimum set of knowledge that one should have as a future scientist or engineer involved in nuclear waste and nuclear energy.

International Workshop on Siting of Radioactive Waste Facilities Nuclear Waste Technical Review Board

Idaho Falls

2023

The mission of the U.S. Nuclear Waste Technical Review Board is to perform unbiased and ongoing technical and scientific peer review of United States Department of Energy nuclear waste management activities. The NWTRB makes an essential contribution to increasing confidence in the scientific process and to informing, from a technical and scientific perspective, decisions on nuclear waste management. The NWTRB provides objective technical information to Congress, the Administration, DOE, government and non-government organizations, and the public on a wide-range of issues related to spent nuclear fuel and high-level waste management and disposition.

National Science Foundation Grant Development Conference

Los Angeles

2019

Portland

2016

Key officials representing program directorates, administrative offices, Office of General Counsel, and Office of the Inspector General will participate in this two-day conference. The conference is considered a must, particularly for new faculty, researchers, educators and administrators who want to gain insight into a wide range of important and timely issues including: the state of current funding; the proposal and award process; and current and recently updated policies.

Collaborative Open Source Curriculum Development Workshop

University of Illinois Urbana – Champaign

2017; 2018

This workshop concluded a year of faculty interaction at six universities to develop curricula for courses common across the same disciplines at multiple universities in order to reduce the amount of time that any individual professor spends on creating what is essentially duplicate materials. The method proposed in this workshop is based on open source software development, where code is shared in online repositories, reviewed by peers, and contributed to the main project.

PyNE Summit

University of Illinois Urbana – Champaign

2017

PyNE is a suite of tools to aid in computational nuclear science and engineering. PyNE seeks to provide native implementations of common nuclear algorithms, as well as Python bindings and I/O support for other industry standard nuclear codes.

Cyber Security Course for Safeguards Practitioners

Pacific Northwest National Laboratory

2018

The course is designed for early to mid-career safeguards practitioners (technical instrument developers, instrument users, policy advisors, etc.) who will benefit from a greater understanding of cyber security threats and how to reduce risks to safeguards systems and processes. This 3.5 day course is designed to teach safeguards and cybersecurity experts how to recognize and mitigate potential cybersecurity vulnerabilities in emerging safeguards instrumentation, information systems, and conduct of operations. This training features classroom style learning opportunities through hands-on exercises and provides plenty of time for questions and discussion. Participants will learn new cyber security skills, use cyber security tools, and collaborate with one another and cyber experts to resolve challenges. They will gain an understanding of cyber security principles and a better awareness of cyber risks associated with safeguards systems. Exercises will include puzzles, exploits of attended and unattended monitoring systems, blended physical and cyber attacks on fictional nuclear facilities, and network defense.

Safeguards Laboratory Day

Pacific Northwest National Laboratory

2017–19

Students and faculty from the University of Idaho invited to Pacific Northwest National Laboratory to learn about the research activities at the laboratory. The day also included hands-on safeguards and security experiments conducted at laboratory facilities, such as materials accounting and vehicle searches.

Next Generation Safeguards Initiative Summer Course

Pacific Northwest National Laboratory

2016

This course, offered through the DOE/NNSA Next Generation Safeguards Initiative, covers major international

safeguards procedures currently in use in IAEA member nations. Daily lectures were supplemented with hands-on safeguards activities conducted by IAEA safeguards inspectors and researchers. Participants included faculty, postdoctorate researchers, and graduate students.

Collaborative Research Planning Meeting

Idaho National Laboratory

2017; 2018

The Critical Materials and Energy Systems Innovation Center is a catalyst for collaborative projects focused on energy research, achieved via connections between the CSIC entities — Idaho National Laboratory, Boise State University, Idaho State University, and the University of Idaho. Through a series of strategic planning meetings, CSIC leadership aims to develop a set of focused research directions. The meeting is intended to establish new collaboration, along with a list of prioritized goals and actions items that will steer internal research investments with the intent of growing sustainable, externally-funded programs. The planning meeting will focus on a strategic area tied to one or more Idaho National Laboratory directorates and will bring together the appropriate Idaho National Laboratory and university leadership and researchers.

Intermountain Energy Summit

Idaho Falls Post Register

2015–17

This summit is held annually and covers energy issues unique to the intermountain region. Participants include faculty from local universities, researchers from national laboratories, energy companies, and politicians. This year, the theme is energy security with a focus on nuclear, renewable, and alternative energy sources and continued advancements in grid and cybersecurity.

United States Department of Energy Collaboration-Based Siting Public Meeting

Boise

2016

The United States Department of Energy is in the initial phase of developing a consent-based process for siting the facilities needed to store and dispose of the nation's spent nuclear fuel and high-level radioactive waste. A consent-based approach to siting relies on understanding the views of the public, stakeholders, and governments at the local, state, and tribal levels. In this first phase, DOE is engaging with interested groups and individuals to learn about what elements are important to consider in designing an enduring approach to siting. This session is an opportunity for the public to share thoughts and perspectives through a facilitated discussion.

Trilateral Nuclear Energy Dialogue: Korea, Japan, United States Global American Business Institute

Boise

2016

A high-level private workshop among preeminent Korean, Japanese, and American experts in nuclear energy and nuclear policy issues, with the intention of fostering relationships, confidence building, and seeking potential areas for trilateral cooperation. In keeping with the overarching theme of previous discussions—the back-end fuel cycle—this meeting seeks to underscore the role of advanced nuclear energy and fuel cycle technologies. Although the obstacles impeding permanent solutions to spent fuel and radioactive waste are largely political, this dialogue aims to highlight the potential of cutting-edge technologies in addressing the policy, environmental, and public acceptance challenges facing management of the nuclear fuel cycle, in addition to opportunities for international collaboration in the research, development, demonstration, and deployment of such technologies.

Idaho's Role in Nuclear: Clean Energy Powered by the Next Generation

Boise State University

2016

Boise State University invited globally recognized leaders in nuclear energy to address the benefits and challenges associated with nuclear energy production and its role in supplying clean energy for a growing world. The summit will also include a panel discussion with John Kotek, Assistant Director of Nuclear Energy, U.S. United States Department of Energy; Dr. Mark Peters, Laboratory Director for the Idaho National Laboratory; Mark Rudin, Vice President for Research at Boise State University; Mike McGough, Chief Commercial Officer at NuScale Power, and Dr. Harold Blackman, Associate Vice President for Research and Economic Development at Boise State University. The panel will address questions about the benefits of and concerns around nuclear power.

Proliferation Resistance & Physical Protection Working Group Workshop

University of California – Berkeley

2015

The PR&PP methodology was developed within the Generation IV International Forum to provide a structured framework to assess the proliferation resistance and physical protection robustness of advanced nuclear energy systems, and to guide designers to further improve their systems. This workshop is intended to familiarize non-experts in this field with the broad aspects of the methodology and its applications. The PR&PP Working Group will present an overview of the methodology to an audience of students, academics, and members of the community who wish to become more familiar with the methodology. To illustrate the methodology, its application to a hypothetical nuclear energy system will be examined. Workshop participants will be divided into subgroups that will consider different proliferation and security threats, and will identify important design features and approaches that contribute to the system's resilience to these threats. Following these subgroup sessions, workshop participants will reconvene to review insights from the subgroups. Finally, an open discussion will be held to obtain feedback from the participants on the approach to PR&PP with the objective of refining the methodology and its presentation to the wider community.

Open Education Resources Development

State of Idaho Board of Education

Boise

2015

This workshop focused on the use of open source educational materials in order to produce an online textbook as a supplement to existing commercial textbooks. An online textbook allows flexibility for faculty to augment content without requiring multiple textbooks for a course. An online textbook allows the educational content of a course to be more closely aligned with the desired learning outcomes. Two online texts have been developed – [Principles of Nuclear Engineering](#) and [Risk Assessment](#).

Advanced Summer School of Radioactive Waste Disposal with Social-Scientific Literacy

University of California – Berkeley

2009; 2011

Hawai'i Tokai International College

2010

This advanced summer school was established in conjunction with the Department of Nuclear Engineering & Management at the University of Tokyo and the Department of Nuclear Engineering at the University of California–Berkeley to provide Ph.D. students and early-career nuclear engineers with education in social sciences and engineering. The goal was to foster a next generation of engineers capable of understanding the public and societal needs, contributing to the societal decision making, and taking a responsible role as engineering experts in society. The discussant leads student group activities by stimulating questions from students and corroborating with the chair to develop a summary of lectures.

Minner Fellows Program

University of California – Berkeley

2011

The objective of this NSF funded program was to develop a framework for making ethical judgments in engineering. Because engineering faculty and graduate students play a leadership role in the development of these technologies, it is essential that they become aware the ethical, legal and social ramifications of them. The course focused on context, or the embodiment of moral maturity and ethical expertise, in the same way that faculty and graduate students embody engineering and technical expertise.

Summer Institute for Preparing Future Faculty

University of California – Berkeley

2005

This unique program is for Doctoral Candidates with an interest in an academic career. Many aspects of teaching are covered: course design, syllabus development, teaching and learning assessment, teaching and learning strategies, and the creation of a teaching portfolio. The program exposes candidates to faculty in several disciplines both within and outside the university; thus allowing for the dissemination of the full scope of teaching methods and skills, as well as broadening of perspectives with regards to the entire teaching profession.

Webinars & Virtual

American Nuclear Society

<i>Equitable Outreach: Now Comes the Hard Part</i>	2025
<i>Next Generation Workforce Development for Nuclear Waste Disposal</i>	2024
<i>Spending Time on Spent Nuclear Fuel with the United States Department of Energy</i>	2024

Idaho Section of the American Nuclear Society

<i>Allies for Career Success</i>	2024
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Boise State University Energy Policy Institute Power Talks

<i>Views from Former Nuclear Regulatory Commission Chairs on Siting for Used Nuclear Fuel Storage</i>	2024
<i>Decision-making and Engagement for Used Nuclear Fuel & Nuclear Waste</i>	2024
<i>Energy Storage</i>	2023
<i>Wildfire-Grid 2.0</i>	2023
<i>Tribal Energy Decisionmaking</i>	2023
<i>Consent-based Siting in Nuclear Energy</i>	2022
<i>Economic Opportunities and Challenges for Idaho with Low Carbon Energy</i>	2022

Western Services Corporation

<i>Western Services Corporation Generation III Generic Pressurized Water Reactor Simulator Workshop</i>	2022
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Nuclear Energy Institute

<i>NEI Advanced Reactors Safeguards & Security Workshop</i>	2021
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University of Idaho

<i>Best Practices for Working with the DOD</i>	2021; 2023
<i>Defense Established Program to Stimulate Competitive Research</i>	2021
<i>NSF Broader Impacts 101 Workshop</i>	2021

HONORS AND AWARDS

University of Idaho American Nuclear Society Student Section

<i>Samuel Glasstone Award for Public Service – Second Place</i>	2019
<i>Certificate of Distinction</i>	2019–

University of California –Berkeley

<i>Nuclear Engineering Department Block Grant Fellowship</i>	2005
<i>Outstanding Graduate Student Instructor Award</i>	2003
<i>Katherina S. DeSharton Fellowship</i>	1999
<i>Hamilton Family Memorial Fellowship</i>	1999

United States Department of Energy · Office of Civilian Radioactive Waste Management

<i>Civilian Radioactive Waste Management Fellowship</i>	2000–04
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Nuclear Energy Institute

<i>National Academy for Nuclear Training Fellowship</i>	1999
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